PS 2010: 4. Probability

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- Homework 2 Due today
- Homework 3 is out
  Due Oct 3th 9:29:59am

What we have learned

- Instantaneous rate of change
- Limit
- Numerical differentiation and derivatives
- Integral: area problem

Today's Agenda

- Probability
- Counting Rules
- Conditional Probability
- Probability mass/density function

What is probability

- **Probability:** is a measure of uncertainty
- Frequentist: the limit of relative frequency.

$$P = \frac{\#\ of\ event\ occurs}{\#\ of\ trials\ in\ repeated\ trials\ under\ the\ same\ conditions}$$

  - Same conditions?
  - Repeated trials?
- Bayesian: a measure of one's subjective belief about the likelihood of an event occurring
  - Too subjective?

Key Concepts

- We need to three concepts to define probability
  - Experiment: actions that produce stochastic events
  - Sample space: a set of all possible outcomes
  - Event: a subset of the sample space
- Example:
  - Flipping a coin
  - Voting

Probability Axioms

- $P(A) >= 0$
- $P(\textit{All possible outcomes}) = 1$
- If events A and B are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$
- Voting example

Venn Diagram

Figure: Event A and B are mutually exclusive

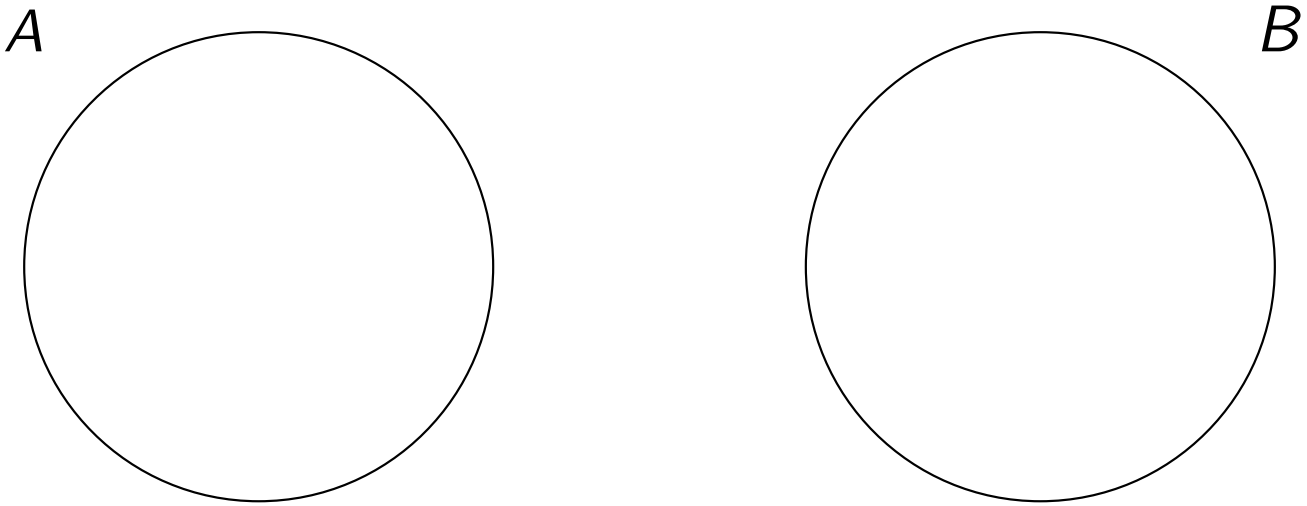
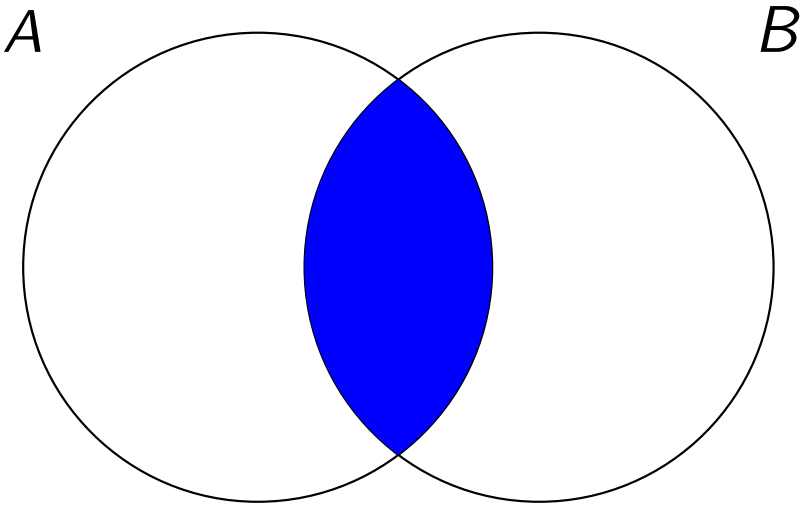


Figure: Event A and B are not mutually exclusive



Additional Rule

- For any events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- *Complement of a set* consist all elements in the sample space except those in the set
  - $P(A^c) = 1 - P(A)$
  - $P(A \cup B) = P(A \cap B^c) + P(B \cap A^c) + P(A \cap B)$
- Law of total probability

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

mn Rule

- With m elements and n elements, it is possible to form $m * n$ pairs containing one element from each group
- Examples:
  - Define # of sample points for tossing two dice
  - What is the probability of different birthday of 20 people from 365 days?

Permutation

- Permutation is a counting technique
- The number of ways in which objects can be arranged
  - Arrange item A,B,C
- Position matters
- When arranging k objects out of n unique objects:

$$P(n,k) = \frac{n!}{(n-k)!}$$

where ! is factorial operator,
$n! = n * (n-1) * (n-2) \cdots * 2 * 1$, and $0! = 1$

Example

- Arrange item A,B,C
- 20 faculty for a committee of 3 people: chair, m1, m2
- Award 2 people from 10 people

Sampling With/Without Replacement

- Sample with replacement: same unit can be repeatedly sampled
- Sample without replacement: each unit can be sampled at most once
- The above example again

Combinations

- When choosing k distinct elements from n elements:

$$ {}_nC_k = \left(\begin{array}{c} n \\ k \end{array} \right) = \frac{P(n,k)}{k!} = \frac{n!}{k!(n-k)!} $$

- Position does not matter
- Examples
  - 20 faculty for a committee of 3 people
  - Award 2 people from 10 people

Example

- A group of three undergraduate and five graduate students are available to fill certain student government posts. If four students are to be randomly selected from this group, find the probability that exactly two undergraduates will be among the four chosen.

Conditional Probability

- Conditional probability: $P(A \mid B)$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A \cap B)$ is the **joint probability**
- $P(B)$ is the **marginal probability**
- We can also have

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

- Conditional probability brings in new information
  - $P(Trump) = 0.49$
  - $P(Trump \mid D)$

Example

Recent polls suggest that
83% of Democrats support impeachment of Trump
44% of Independents support impeachment of Trump
14% of Republicans support impeachment of Trump
Meanwhile, $P(D) = 31\%$, $P(I) = 40\%$, $P(R) = 29\%$

- What is the average probability of supporting for impeachment?
- Is this true for all Americans?
- Find the probability that a randomly selected American is a Democrat who supports impeachment.
- Now suppose that the randomly selected American supports impeachment. How does this information change the probability that the selected American belongs to a particular political party?

Independence

- Knowledge of one event does not provide any more information about the other event

$$P(A \mid B) = P(A) \text{ and } P(B \mid A) = P(B)$$

- If two events A and B are independent:

$$P(A \cap B) = P(A)P(B)$$

- Example: randomization

Bayes Rule

- Update our beliefs after we observe the data
- **Definition:**

$$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A)+P(B|A^c)P(A^c)}$$

  $P(A)$ is prior probability, $P(A \mid B)$ is posterior probability
- Recall law of total probability is

$$P(B) = P(A \cap B) + P(B \cap A^c)$$

  we know
- $P(A \cap B) = P(B \mid A)P(A)$
- Therefore, $P(B) = P(B \mid A)P(A) + P\left(B \mid A^c\right)P\left(A^c\right)$

Example

The fraction of persons in the population who have a particular disease is .01. The probability that a particular diagnostic test gives a positive results is .2. The probability that the test gives a positive result given that an individual actually has the disease is .8. If an individual has the test done, and it comes back positive, what is the probability that she or he has the disease? Why is this probability not equal to one?

Example

In a certain assembly plant, three machines, B1, B2, and B3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective.

- Suppose that a finished product is randomly selected. What is the probability that it is defective?
- If a product was chosen randomly and found to be defective, what is the probability that it was made by machine B3?

Random Variable

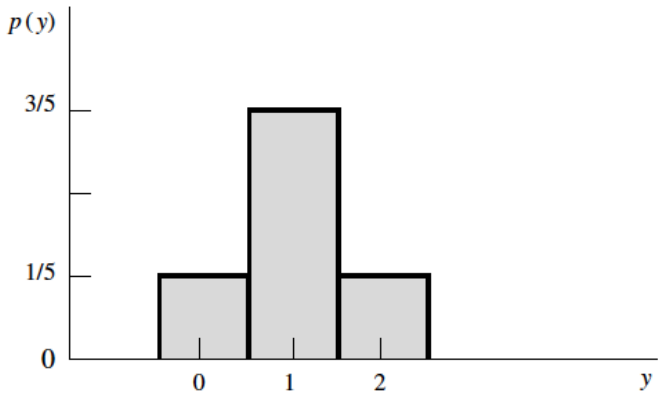
- **Random Variable** assigns a numeric value to each event of the experiment.
- Values need to be mutually exclusive and exhaustive events
- Two types of random variables:
  - Discrete random variable: takes a finite or at most countably infinite number of distinct values
  - Continuous: uncountably infinite number of values.

Probability Distribution for Discrete Random Variable

- Probability that Y takes on the value y is $P(Y = y)$ or $p(y)$
- Probability distribution for a discrete variable provides $p(y) = P(Y = y)$ for all y
- For discrete random variable, it is also **probability mass function, pmf**

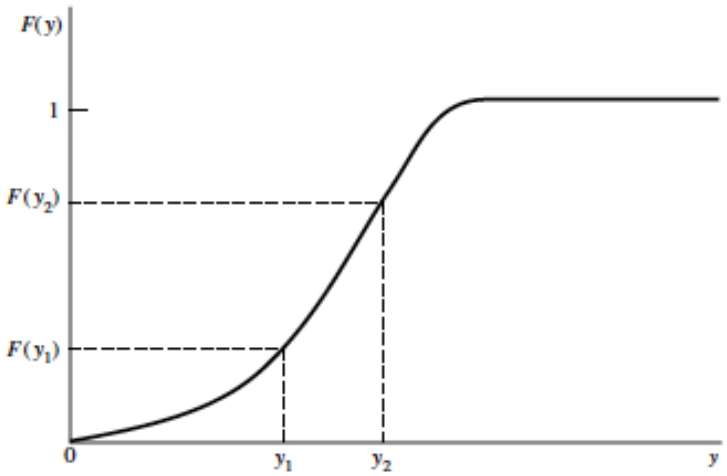
Table 3.1 Probability distribution for Example 3.1

y	$p(y)$
0	1/5
1	3/5
2	1/5



Probability Distribution for Continuous Random Variable

- Probability that Y takes on a specific value y is $P(Y = y) = p(y) = 0$
- Probability distribution for a continuous variable denoted by $F(y)$ is such that $F(y) = P(Y \leq y)$ for all $-\infty < y < \infty$
- The function $p(y)$ is **probability density function, pdf** for continuous variable



Cumulative Distribution Function

- The **cumulative distribution function** $F(y)$ of a discrete random variable y with probability distribution $f(x)$ is
$$F(y) = P(Y \leq y) = \sum_{t \leq y} f(y)$$
- The cumulative distribution function $F(y)$ of a continuous random variable y with probability distribution $f(x)$ is
$$F(y) = P(Y \leq y) = \int_{-\infty}^{y} f(t)dt$$
$$\text{for all } -\infty < y < \infty$$
- $\int_{-\infty}^{\infty} f(x)dx = 1$

$f(x)$ is the probability mass function or probability density function. $f(x) = p(x)$ just different notation.

$$P(a < X < b) = \int_a^b f(x)\, dx.$$

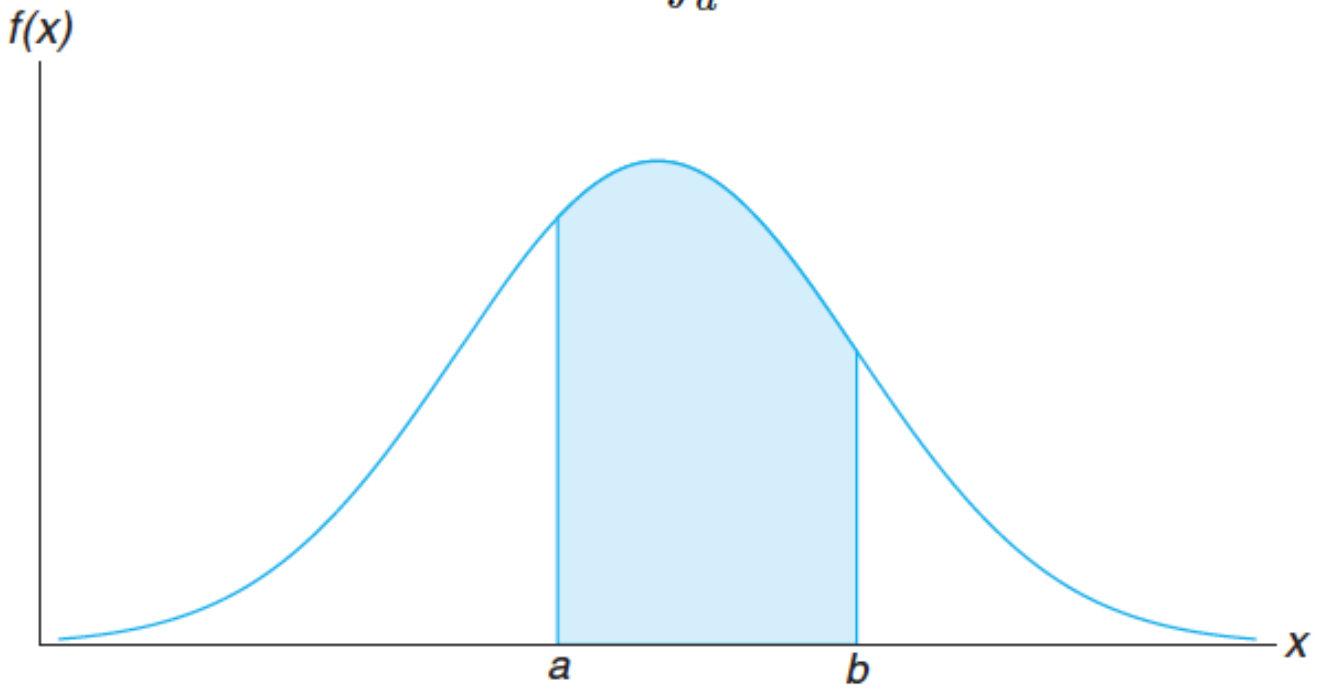


Figure 3.5: $P(a < X < b)$.

Example

If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags.

- Find a formula for the probability distribution of the number of cars with side airbags among the next 4 cars sold by the agency.
- Find the cumulative distribution function of the random variable

Example

For a continuous random variable X, the probability density function is

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

- Find $P(-\infty < X \leq \infty)$
- Find $P(0 < X \leq 1)$

Expectation

- Y is a discrete random variable, then the expected value of Y denoted as $E(Y)$ is $\mu = E(y) = \sum yf(y)$
- If Y is a continuous variable, then $\mu = E(y) = \int_{-\infty}^{\infty} yf(y)$
- Expectation represent the mean of random variables
- Let Y be a random variable with pdf f(y), the expected value of the discrete random variable g(Y) is

$$\mu_{g(Y)} = E[g(Y)] = \sum_y g(y)f(y)$$

- The expected value of the continuous random variable is

$$\mu_{g(Y)} = E[g(Y)] = \int_{-\infty}^{\infty} g(y)f(y)dy$$

Example

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

Example

Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20{,}000}{x^3}, & x > 100 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected life of this type of device.

Example

Suppose that the number of cars X that pass through a car wash between 4 : 00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

Example

Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find the expected value of $g(X) = 4X + 3$.

Variance

Definition: If Y is a random variable with mean $E(Y) = \mu$, the variance of a random variable Y is defined to be the expected value of $(Y - \mu)^2$:

$$V(Y) = E[(Y - \mu)^2]$$

Example: The probability distribution for a random variable Y is given in the table below. Find the mean, variance, and standard deviation of Y.

y	$p(y)$
0	$1/8$
1	$1/4$
2	$3/8$
3	$1/4$